MATHEMATICS / INDEPENDENT STUDY

# MM7030

# Category Theory

Exercises and mock assessments

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**How to use this volume** E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

# 1 Unit 1 · Categories and duality

**MM7030-U1.01 · E · 15 min**
Learning objective: U1.1 Category axioms
Verify that nonnegative integers with an arrow $m \to n$ exactly when $m \leq n$ form a category.

**MM7030-U1.02 · E · 15 min**
Learning objective: U1.1 One-object categories
Describe the one-object category associated with the additive monoid $\mathbb{N}$. What arrows are invertible?

**MM7030-U1.03 · E · 15 min**
Learning objective: U1.2 Opposites
Describe the opposite of a poset category.

**MM7030-U1.04 · E · 15 min**
Learning objective: U1.1 Isomorphisms
Prove an isomorphism in the category of groups is exactly a bijective group homomorphism.

**MM7030-U1.05 · C · 30 min**
Learning objective: U1.2 Monomorphisms
Prove the composite of two monomorphisms is monic.

**MM7030-U1.06 · C · 30 min**
Learning objective: U1.2 Cancellation
If $gf$ is monic, prove $f$ is monic. Must $g$ be monic?

**MM7030-U1.07 · C · 30 min**
Learning objective: U1.2 Split arrows
Show a morphism with a left inverse is monic. Give a monomorphism of sets with no left inverse.

**MM7030-U1.08 · C · 30 min**
Learning objective: U1.2 Epimorphisms
Prove a function missing an element of its codomain is not epic in sets.

**MM7030-U1.09 · C · 30 min**
Learning objective: U1.1 Groupoids
Show the isomorphisms in any category, retaining all objects, form a subcategory in which every arrow is invertible.

**MM7030-U1.10 · A · 50 min**
Learning objective: U1.2 Balancedness
Prove a morphism that is both monic and split epic is an isomorphism.

**MM7030-U1.11 · A · 50 min**
Learning objective: U1.2 Rings versus sets
Prove the inclusion $\mathbb{Z} \to \mathbb{Z}\left[\frac{1}{2}\right]$ is epic among commutative unital rings but is not an isomorphism.

**MM7030-U1.12 · A · 50 min**
Learning objective: U1.1 Categories from paths
Construct the free category on a directed graph and prove any assignment of its vertices and edges to objects and arrows of a category extends uniquely to a functor.

### MM7030-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: Words as arrows

Form a category with one object whose arrows are finite words in the two symbols $a, b$, including the empty word; composition is concatenation. Verify the category laws, determine which arrows are monomorphisms and epimorphisms, and determine which are isomorphisms.

# 2 Unit 2 · Functors and equivalences

**MM7030-U2.01 · E · 15 min**
Learning objective: U2.1 Functor laws
Show a monotone map between posets gives a functor between their categories.

**MM7030-U2.02 · E · 15 min**
Learning objective: U2.1 Forgetful functors
Define the forgetful functor from vector spaces over $K$ to sets on objects and arrows.

**MM7030-U2.03 · E · 15 min**
Learning objective: U2.1 Hom functors
For $f : X \to Y$, write the map $\mathcal{C}(A, f)$.

**MM7030-U2.04 · E · 15 min**
Learning objective: U2.2 Faithfulness
Explain why the forgetful functor from groups to sets is faithful.

**MM7030-U2.05 · C · 30 min**
Learning objective: U2.1 Contravariance
Verify inverse image of subsets is a contravariant functor on sets.

**MM7030-U2.06 · C · 30 min**
Learning objective: U2.2 Fullness
Show the forgetful functor from groups to sets is not full.

**MM7030-U2.07 · C · 30 min**
Learning objective: U2.2 Reflection of isomorphisms
Prove a full and faithful functor reflects isomorphisms.

**MM7030-U2.08 · C · 30 min**
Learning objective: U2.1 One-object functors
Show functors between one-object categories of monoids are precisely monoid homomorphisms.

**MM7030-U2.09 · C · 30 min**
Learning objective: U2.2 Matrix category
Show the category of matrices over $K$, with objects nonnegative integers, is equivalent to finite-dimensional vector spaces.

**MM7030-U2.10 · A · 50 min**
Learning objective: U2.2 Faithful reflection
Prove a faithful functor reflects monomorphisms and epimorphisms, without assuming fullness.

**MM7030-U2.11 · A · 50 min**
Learning objective: U2.2 Equivalence of posets
Show equivalent poset categories are isomorphic as posets.

**MM7030-U2.12 · A · 50 min**
Learning objective: U2.2 Failure of preservation
Construct a faithful functor which sends a monomorphism and epimorphism to a function which is neither. Use the two-object poset $0 < 1$ as source, and explain why this does not contradict U2.10.

### MM7030-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Discarding redundant points

Let objects be pairs $(X, e)$ with $X$ a finite set and $e : X \to X$ satisfying $e^2 = e$. An arrow $f : (X, e) \to (Y, d)$ satisfies $fe = f = df$, and the identity of $(X, e)$ is $e$. Verify this is a category. Show that sending $(X, e)$ to its image $e(X)$ and arrows to restrictions gives an equivalence with finite sets.

# 3 Unit 3 · Natural transformations

**MM7030-U3.01 · E · 15 min**
Learning objective: U3.1 Naturality
Write the naturality equation for $\eta : F \Rightarrow G$ at $f : A \to B$.

**MM7030-U3.02 · E · 15 min**
Learning objective: U3.2 Identities
Show the identity maps $1_{FA}$ form a natural transformation $F \Rightarrow F$.

**MM7030-U3.03 · E · 15 min**
Learning objective: U3.1 Natural projections
Show $\pi_1 : X \times X \to X$ is natural in the set $X$.

**MM7030-U3.04 · E · 15 min**
Learning objective: U3.2 Components
Describe a natural transformation between two constant functors with values $A, B$ when the source is the terminal one-object category.

**MM7030-U3.05 · C · 30 min**
Learning objective: U3.1 Diagonal
Show the diagonal $\Delta_X : X \to X \times X$ is natural and each component is injective.

**MM7030-U3.06 · C · 30 min**
Learning objective: U3.2 Vertical composition
Prove vertical composition of natural transformations is associative.

**MM7030-U3.07 · C · 30 min**
Learning objective: U3.1 Inverse naturality
If $\eta : F \Rightarrow G$ is natural and each component is invertible, prove the inverse family is natural.

**MM7030-U3.08 · C · 30 min**
Learning objective: U3.2 Group actions
For two actions of a group $G$ on sets $X, Y$, show natural transformations of the corresponding functors are exactly equivariant functions.

**MM7030-U3.09 · C · 30 min**
Learning objective: U3.1 Double dual
Verify naturality of the canonical map $V \to V^{**}$ by evaluating on a linear functional.

**MM7030-U3.10 · A · 50 min**
Learning objective: U3.1 Natural endomorphisms of sets
Classify all natural transformations from the identity functor on sets to itself without using Yoneda.

**MM7030-U3.11 · A · 50 min**
Learning objective: U3.2 Center of a group
Identify the natural endomorphisms of the identity functor of a group regarded as a one-object category.

**MM7030-U3.12 · A · 50 min**
Learning objective: U3.1 Scalar natural operations (proof reconstruction)
Show every natural endomorphism of the identity functor on $K$-vector spaces is multiplication by a single scalar.

### MM7030-U3.13 · A · 60 min · Optional challenge

Learning objective: U3 Transfer: Uniform operations on subsets

Let $P$ send a set to its power set and a function to direct image. Classify all natural transformations $P \Rightarrow P$ on the category of sets.

# 4 Unit 4 · Universal properties

**MM7030-U4.01 · E · 15 min**
Learning objective: U4.1 Initial and terminal
Identify initial and terminal objects in sets and in groups.

**MM7030-U4.02 · E · 15 min**
Learning objective: U4.1 Products
Describe the product of a two-element set and a three-element set with its projections.

**MM7030-U4.03 · E · 15 min**
Learning objective: U4.2 Coproducts
Why is ordinary union not always a coproduct of two sets?

**MM7030-U4.04 · E · 15 min**
Learning objective: U4.1 Poset products
Find the product and coproduct of twelve and eighteen in positive integers ordered by divisibility.

**MM7030-U4.05 · C · 30 min**
Learning objective: U4.1 Product uniqueness
Prove two products of fixed objects are uniquely isomorphic compatibly with their projections.

**MM7030-U4.06 · C · 30 min**
Learning objective: U4.2 Finite biproducts
Prove $V \oplus W$ is both a product and a coproduct in vector spaces.

**MM7030-U4.07 · C · 30 min**
Learning objective: U4.2 Free vector spaces
Construct the linear extension of a function from $\{a, b\}$ into a vector space, taking $a$ to $u$ and $b$ to $v$.

**MM7030-U4.08 · C · 30 min**
Learning objective: U4.1 Terminal products
Show $A \times 1$ is isomorphic to $A$ whenever $1$ is terminal and the product exists.

**MM7030-U4.09 · C · 30 min**
Learning objective: U4.2 Groups and coproducts
Explain why $C_2 \times C_2$ with its coordinate inclusions is not a coproduct of two copies of $C_2$ in groups.

**MM7030-U4.10 · A · 50 min**
Learning objective: U4.1 Product associativity
Construct the canonical isomorphism $(A \times B) \times C \to A \times (B \times C)$ solely from universal properties.

**MM7030-U4.11 · A · 50 min**
Learning objective: U4.2 Infinite sums and products
For countably many copies of $K$, show the finite-support direct sum is a coproduct but generally not the product under its coordinate projections.

**MM7030-U4.12 · A · 50 min**
Learning objective: U4.1 Joint detection by projections
Prove product projections jointly determine every arrow into the product. Deduce the naturality of the canonical swap $A \times B \to B \times A$ in both variables.

### MM7030-U4.13 · A · 60 min · Optional challenge

Learning objective: U4 Transfer: Joining pointed sets

In the category of pointed sets and basepoint-preserving functions, construct the coproduct of two finite pointed sets of sizes $m$, $n$. Prove its universal property, compute its size, and compare it with their product.

# 5 Unit 5 · Limits and colimits

**MM7030-U5.01 · E · 15 min**
Learning objective: U5.1 Equalizers
Find the equalizer in sets of $f, g : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n^2, g(n) = n$.

**MM7030-U5.02 · E · 15 min**
Learning objective: U5.1 Pullbacks
Find the pullback of $f : \mathbb{Z} \to \mathbb{Z}$, $f(a) = 2a$, and $g : \mathbb{Z} \to \mathbb{Z}$, $g(b) = 3b$, in sets.

**MM7030-U5.03 · E · 15 min**
Learning objective: U5.2 Coequalizers
In sets, coequalize the two maps from a singleton to $\{a, b, c\}$ selecting $a$ and $b$.

**MM7030-U5.04 · E · 15 min**
Learning objective: U5.1 Empty diagrams
Explain why a limit of the empty diagram is terminal and a colimit is initial.

**MM7030-U5.05 · C · 30 min**
Learning objective: U5.1 Equalizers are monic
Prove an equalizer arrow is monic using only its universal property.

**MM7030-U5.06 · C · 30 min**
Learning objective: U5.1 Pullbacks of monos
Prove a pullback of a monomorphism is monic.

**MM7030-U5.07 · C · 30 min**
Learning objective: U5.2 Linear coequalizers
Prove the coequalizer of linear maps $f, g : V \to W$ is $W/\operatorname{im}(f-g)$.

**MM7030-U5.08 · C · 30 min**
Learning objective: U5.2 Pushout of sets
Compute the pushout of inclusions $\{b\} \to \{a, b\}$ and $\{b\} \to \{b, c\}$ in sets.

**MM7030-U5.09 · C · 30 min**
Learning objective: U5.1 Limits in groups
Show the equalizer of group homomorphisms $f, g : G \to H$ is a subgroup of $G$. Must it be normal?

**MM7030-U5.10 · A · 50 min**
Learning objective: U5.1 Finite-limit construction
Construct the pullback of $f : A \to C, g : B \to C$ using a product and an equalizer.

**MM7030-U5.11 · A · 50 min**
Learning objective: U5.2 Gluing may identify more
Give a coequalizer in sets where the initially required identification pairs are not themselves transitive, and compute the actual quotient.

**MM7030-U5.12 · A · 50 min**
Learning objective: U5.1 Pullback pasting
Prove that two horizontally composable pullback squares have a pullback as their outer rectangle.

### MM7030-U5.13 · A · 60 min · Optional challenge

Learning objective: U5 Transfer: Compatible integer measurements

Fix a positive integer $n$. Find an abelian group $P$ and maps $p, q : P \to \mathbb{Z}$ with $p$ congruent to $2q$ modulo $n$, such that every pair $f, g : A \to \mathbb{Z}$ with this congruence factors uniquely through $p, q$. Give a basis for $P$.

# 6 Unit 6 · Representability and Yoneda

**MM7030-U6.01 · E · 15 min**
Learning objective: U6.1 Yoneda evaluation
For $\eta : \mathcal{C}(A, -) \Rightarrow F$, which element determines $\eta$ under Yoneda?

**MM7030-U6.02 · E · 15 min**
Learning objective: U6.2 Representing the identity
Show the identity functor on sets is covariantly represented by a singleton.

**MM7030-U6.03 · E · 15 min**
Learning objective: U6.1 Reconstructing a transformation
Let $x \in FA$. Define the corresponding transformation from $\mathcal{C}(A, -)$ to $F$.

**MM7030-U6.04 · E · 15 min**
Learning objective: U6.2 Universal elements
For the underlying-set functor on groups represented by $\mathbb{Z}$, identify a universal element.

**MM7030-U6.05 · C · 30 min**
Learning objective: U6.1 Natural binary operations
Classify natural transformations $X \times X \to X$ in sets.

**MM7030-U6.06 · C · 30 min**
Learning objective: U6.2 Contravariant representables
Given $f : A \to B$, write the natural transformation $\mathcal{C}(-, A) \Rightarrow \mathcal{C}(-, B)$ and prove naturality.

**MM7030-U6.07 · C · 30 min**
Learning objective: U6.2 Universal element category
Show a universal element of a covariant functor is an initial object in its category of elements.

**MM7030-U6.08 · C · 30 min**
Learning objective: U6.1 Empty representing set
Classify natural transformations from the constant singleton functor on sets to the identity functor.

**MM7030-U6.09 · C · 30 min**
Learning objective: U6.2 Uniqueness
Prove that naturally isomorphic contravariant representables have isomorphic representing objects.

**MM7030-U6.10 · A · 50 min**
Learning objective: U6.1 Natural power operations on groups
Classify all natural transformations from the underlying-set functor on groups to itself.

**MM7030-U6.11 · A · 50 min**
Learning objective: U6.2 Nonrepresentability
Show the constant empty-set functor from sets to sets is not covariantly representable.

**MM7030-U6.12 · A · 50 min**
Learning objective: U6.2 Natural n-ary operations
For finite sets $m, n$, classify natural transformations $X^n \to X^m$ on sets, including the empty cases.

### MM7030-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: Families of subsets from pairs

For a function $f : X \to Y$, define $P(f)(S) = f[S]$ and $P(P(f))(\mathcal{A}) = \{f[S] : S \in \mathcal{A}\}$. Classify all families $\theta_X : X \times X \to P(P(X))$ satisfying $P(P(f))\left(\theta_{X(x,y)}\right) = \theta_{Y(f(x),f(y))}$ for every $f$. How many such families are there?

# 7 Unit 7 · Adjunctions

**MM7030-U7.01 · E · 15 min**
Learning objective: U7.1 Adjunction types
For $F : \mathcal{C} \to \mathcal{D}$ left adjoint to $G$, write the hom-set bijection and the types of the unit and counit components.

**MM7030-U7.02 · E · 15 min**
Learning objective: U7.1 Currying
Curry the function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(b, a) = 2b - a$.

**MM7030-U7.03 · E · 15 min**
Learning objective: U7.2 Evaluation
Describe the counit of the product–exponential adjunction in sets.

**MM7030-U7.04 · E · 15 min**
Learning objective: U7.2 Free-vector-space unit
Describe the unit of the free-vector-space/forgetful adjunction.

**MM7030-U7.05 · C · 30 min**
Learning objective: U7.1 Power-set adjunction
Prove direct image along $f : X \to Y$ is left adjoint to inverse image on power-set posets.

**MM7030-U7.06 · C · 30 min**
Learning objective: U7.1 Floor and inclusion
Let $i : \mathbb{Z} \to \mathbb{R}$ be inclusion as ordered sets. Show $i$ is left adjoint to floor.

**MM7030-U7.07 · C · 30 min**
Learning objective: U7.2 Free counit
For a vector space $V$, describe the counit $FUV \to V$ and verify the triangle involving the unit on $UV$.

**MM7030-U7.08 · C · 30 min**
Learning objective: U7.2 Transpose formulas
Given unit and counit, write the transpose of $h : FC \to D$ and the inverse transpose of $k : C \to GD$.

**MM7030-U7.09 · C · 30 min**
Learning objective: U7.1 Diagonal adjunctions
Show the diagonal functor $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ is left adjoint to the binary-product functor when products exist.

**MM7030-U7.10 · A · 50 min**
Learning objective: U7.2 Triangle converse
Prove the two unit–counit transpose formulas are inverse assuming naturality and the triangle identities.

**MM7030-U7.11 · A · 50 min**
Learning objective: U7.1 Composite adjunctions
If $F \dashv G$ and $F' \dashv G'$ are composable adjunctions, prove $F'F \dashv \mathbb{G}'$.

**MM7030-U7.12 · A · 50 min**
Learning objective: U7.2 Closure operators
For an adjunction $F : P \to Q$, $G : Q \to P$ of posets, prove $GF$ is monotone, extensive, and idempotent.

### MM7030-U7.13 · A · 60 min · Optional challenge

Learning objective: U7 Transfer: A relation produces a closure

Given a relation $R \subset X \times Y$, let $E(A)$ be the set of points of $Y$ related to some member of $A \subset X$. For $B \subset Y$, find the greatest subset $U(B) \subset X$ with $E(U(B)) \subset B$. Prove $A \subset U(E(A))$ and $U(E(U(E(A)))) = U(E(A))$.

# 8 Unit 8 · Preservation and applications

**MM7030-U8.01 · E · 15 min**
Learning objective: U8.1 Preservation direction
Which of products and coproducts must a right adjoint preserve, when they exist?

**MM7030-U8.02 · E · 15 min**
Learning objective: U8.1 Terminal objects
Explain why a right adjoint preserves terminal objects.

**MM7030-U8.03 · E · 15 min**
Learning objective: U8.2 Abelianization
What is the abelianization of an abelian group?

**MM7030-U8.04 · E · 15 min**
Learning objective: U8.2 Exponentials
In sets, how many functions are in $B^A$ if $|A| = 3$ and $|B| = 2$? Explain its role in currying.

**MM7030-U8.05 · C · 30 min**
Learning objective: U8.1 Representable preservation
Show $\mathcal{C}(A, -)$ preserves a binary product.

**MM7030-U8.06 · C · 30 min**
Learning objective: U8.1 Nonpreservation
Show the free-vector-space functor does not preserve the terminal set.

**MM7030-U8.07 · C · 30 min**
Learning objective: U8.2 Abelianization map
Compute the abelianization of $S_3$.

**MM7030-U8.08 · C · 30 min**
Learning objective: U8.2 Free coproducts
Construct the canonical isomorphism $F(A \coprod B) \cong F(A) \oplus F(B)$ for free vector spaces.

**MM7030-U8.09 · C · 30 min**
Learning objective: U8.2 Implication in a poset
For subsets of $S$, prove $X \cap A \subseteq B$ iff $X \subseteq (S \setminus A) \cup B$.

**MM7030-U8.10 · A · 50 min**
Learning objective: U8.1 Right-adjoint preservation
Prove directly through adjunct transposes that a right adjoint preserves an equalizer.

**MM7030-U8.11 · A · 50 min**
Learning objective: U8.2 Abelianization adjunction
Prove abelianization is left adjoint to inclusion of abelian groups, including naturality.

**MM7030-U8.12 · A · 50 min**
Learning objective: U8.1 Equivalences preserve limits
Prove an equivalence of categories preserves all existing small limits.

### MM7030-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: A quotient that loses a kernel

On abelian groups define $Q(A) = A/2A$, with induced maps. Prove that $Q$ preserves coproducts and coequalizers, but does not preserve all kernels.

# 9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

**MM7030-M1.01 · E · 15 min**
Learning objective: U1 Cancellation
State the definition of epimorphism and prove every surjective set function is epic.

**MM7030-M1.02 · E · 15 min**
Learning objective: U4 Universal properties
State the coproduct property and identify the coproduct of two finite sets of sizes three and four.

**MM7030-M1.03 · C · 30 min**
Learning objective: U3 Naturality
Prove the swap maps $X \times X \to X \times X$, $(x, y) \mapsto (y, x)$ form a natural isomorphism.

**MM7030-M1.04 · C · 30 min**
Learning objective: U5 Limits
Construct the pullback in sets of two maps to a singleton and prove its universal property.

**MM7030-M1.05 · A · 40 min**
Learning objective: U6 Yoneda
Classify natural transformations $X^3 \to X^2$ in sets and count them.

**MM7030-M1.06 · A · 40 min**
Learning objective: U7–U8 Adjunctions
Prove a left adjoint preserves initial objects directly from its hom-set bijection.

# 10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

**MM7030-M2.01 · E · 15 min**
Learning objective: U2 Functors
Prove every functor preserves isomorphisms.

**MM7030-M2.02 · E · 15 min**
Learning objective: U7 Unit and counit
For free vector spaces, describe the unit at a set $X$ and counit at a vector space $V$.

**MM7030-M2.03 · C · 30 min**
Learning objective: U5 Coequalizers
Compute the coequalizer in vector spaces of $f, g : K \to K^2$ given by $f(t) = (t, 0)$ and $g(t) = (0, t)$.

**MM7030-M2.04 · C · 30 min**
Learning objective: U6 Representability
Show the functor $G \mapsto G \times G$ from groups to sets is represented by a free group on two generators.

**MM7030-M2.05 · A · 40 min**
Learning objective: U2–U5 Equivalence
Let $F$ be full, faithful, and essentially surjective. Prove it sends a terminal object to a terminal object.

**MM7030-M2.06 · A · 40 min**
Learning objective: U7–U8 Logic
In the power-set poset of $S$, identify a right adjoint to intersection with a fixed $A$ and derive its counit inequality.